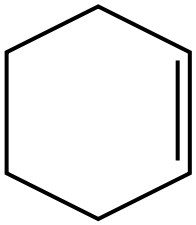
**7.b**

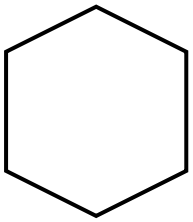
**i)** $\underbrace{C_3H_8}_{propane} + 5O_2 \rightarrow 3CO_2 + 4H_2O$

**ii)** $\underbrace{H-\overset{H}{\overset{|}{C}}=\overset{H}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-H}_{propene} + \underbrace{H_2O}_{Steam} \overset{H_3PO_4/SiO_2}{\rightleftharpoons} \underbrace{H-\overset{H}{\overset{|}{C}}-\overset{OH}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-H}_{propan-2-ol}$

**iii)**



$+ H_2 \overset{Pd/C}{\rightleftharpoons}$



Cyclohexene Cyclohexane

**7.c**

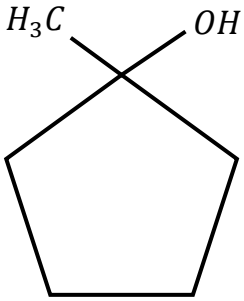
**i)** $H-\overset{H}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-\overset{OH}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-H$

3 - pentanol (Secondary)

**ii)** $H-\overset{H}{\overset{|}{C}}-\overset{CH_3}{\overset{|}{C}}-\overset{H}{\overset{|}{C}}-OH$

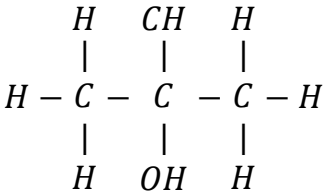
2, 2 – dimethyl – 1 – propanol (Primary)

**iii)**




1 – methylcycopentanol (Tertiary)

**iv)**



2 – methyl – 2 – propanol (Tertiary)

**8.a**

**i)**

| ***Element*** | ***C*** | ***H*** | ***O*** |
|---|---|---|---|
| % of mass | 40 | 6.7 | 53.3 |
| No of mole | $\frac{40}{12} = 3.33$ | $\frac{6.7}{1} = 6.7$ | $\frac{53.3}{16} = 3.33$ |
| Mole ratio | $\frac{3.33}{3.33} = 1$ | $\frac{6.7}{3.33} = 2$ | $\frac{6.7}{3.33} = 1$ |

$Empirical\ formula\ is\ CH_2O$

**ii)** $Molecular\ formular\ of\ x$

$Relative\ molecular\ mass\ of\ x = 2 \times Vapour\ density$

$= 2 \times 30 = 60$

$Molecular\ formular = (Empirical\ formular)_n$

$[CH_2O]_n = 60$

$[12 + 2 + 16]_n = 60$

$30n = 60$

$n = \frac{60}{30} = 2$

$Molecular\ formula\ is\ C_2H_4O_2$

**8.b** Carbondioxide

**8.c**

$$-\overset{\overset{O}{\|}}{C}-O-H$$

**8.d**

$$H-\overset{\overset{H}{|}}{\underset{\underset{H}{|}}{C}}-\overset{\overset{O}{\|}}{C}-O-H$$

**8.e** Esterification reaction

$$\underbrace{H_3C-\overset{\overset{O}{\|}}{C}-O-H}_{ethanoic\ acid} + \underbrace{CH_3CH_2-OH}_{ethanol} \rightarrow \underbrace{H_3C-\overset{\overset{O}{\|}}{C}-O\ CH_2CH_3}_{ethyl\ ethanoate} + H_2O$$

Detailed solutions to

# JUPEB

# 2018

# CHEMISTRY

Past Question and Answer
with detailed explanations
and solutions.



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## SECTION A: MULTIPLE CHOICE QUESTIONS

**Answer ALL Questions**

**1.** The electronic configuration for $Cu$ is

**A.** $1s^2 2s^2 2p^6 3s^2 3p^6$

**B.** $1s^2 2s^2 2p^6 3s^2 3p^6 4s^2 3d^9$

**C.** $1s^2 2s^2 2p^6 3s^2 3p^6 4s^1 3d^{10}$

**D.** $1s^2 2s^2 3s^2 2p^6 4s^2 2p^6$

**2.** What is the value of angular momentum quantum of a $3d$ orbital?

**A.** 3

**B.** 0

**C.** 1

**D.** 2

**3.** The energy needed to remove an electron in the outermost shell of a gaseous atom is called?

**A.** Ionization energy

**B.** Electron affinity

**C.** Electrode potential

**D.** Photoelectronic energy

**4.** Which of the following atomic properties generally increases down a group in the periodic table of elements?

**A.** Electronegativity

**B.** Atomic radius

**C.** Electron affinity

**D.** Ionization energy

**5.** Substances that are electron pair acceptors are called

**A.** Arrhenius bases

**B.** Lewis acids

**C.** Bronsted Lowry acids

**D.** Lewis bases

**6.** All the A shampoo solution at $25°C$ has a hydroxide-ion concentration of $1.5 \times 10^{-9} moldm^{-3}$. The solution is

**A.** Acidic

**B.** Neutral

**C.** Basic

**D.** Amphiprotic

**7.** In a measurement, significant figures include

**A.** All digits except zero that are measured

**B.** All digits except zero that are not measured

**C.** All zeros at the beginning of a number

**D.** All digits in a number

**8.** The degeneracy of orbitals with angular momentum quantum number $i = 2$ is

**A.** 2

**B.** 3

**C.** 5

**D.** 1

**9.** What is the principle that allows the use of mass spectrometry in determination of the molecular weight of a compound?

**A.** Higher molecular weight compounds are less volatile

**B.** Higher molecular weight compounds are denser

**C.** A beam of higher molecular weight cations is deflected less by magnetic field

**D.** Ions with higher molecular weight absorb higher frequencies of light

**10.** State What is the oxidation state of oxygen in $OF_2$?

**A.** $+2$

**B.** $-2$

**C.** $-1$

**D.** $+1$

**11.** Which of the following is not true about ionization energy?

**A.** Ionization energy increases across a period

**B.** Reversal in trend of ionization occurs in passing from beryllium to baron

**C.** Half-filled subshell makes ionization energy of oxygen lower than that of nitrogen

**D.** Ionization energy does not depend on nuclear charge

**12.** In terms of Lewis concept, an acid and a base are a/an:

**A.** Proton donor and proton acceptor respectively

**B.** Electron donor and electron acceptor respectively

**C.** Proton acceptor and proton donor respectively

**D.** Electron acceptor and electron donor respectively

**13.** Which of the following will affect the equilibrium position of the reaction?

$I_{2(g)} + H_{2(g)} \rightleftharpoons 2HI_{(g)}$

**A.** Addition of a catalyst

**B.** Increase in pressure

**C.** Decrease in pressure

**D.** Addition of HI

**14.** Which of the following ions has the smallest size?

**A.** $O^{2+}$

**B.** $Na^{+}$

**C.** $Mg^{2+}$

**D.** $F^{-}$

**15.** A sodium pellet is dropped into warm water and $500cm^3$ of hydrogen gas is liberated. What mass of sodium is involved in the reaction? $[Na = 23;\ V_m = 22400cm^3]$

**A.** $1.03g$

**B.** $0.51g$

**C.** $2.30g$

**D.** $0.98g$

**16.** Calculate the percentage by mass of water of crystallization in $MgSO_4.7H_2O$

$[Mg = 24, S = 32, O = 16, H = 1]$

**A.** $51.2\%$

**B.** $56.7\%$

**C.** $50.12\%$

**D.** $52.55\%$

**17.** What volume of $5\ moldm^{-3} tetraoxosulphate\ (VI)\ acid$ is required to prepare $300cm^3\ 0f\ 0.5moldm^{-3}$ solutions?

**A.** $30cm^3$

**B.** $50cm^3$

**C.** $3cm^3$

**D.** $30cm^3$

**18.** An iron complex with molecular formula $C_{34}H_{32}FeN_4O_4$, constitutes $6\%$ of haemoglobin by mass. If a blood sample contains $0.75g$ haemoglobin, what mass of $Fe$ is contributed by the complex?

$[O = 12, H = 1, Fe = 56, N = 14, O = 16]$

**A.** $0.0041g$

**B.** $0.0450g$

**C.** $0.06000g$

**D.** $0.0056g$

**19.** The temperature above which the liquid states of a substance no longer exist is called

**A.** Supercritical temperature

**B.** Critical pressure

**C.** Critical temperature

**D.** Critical point

**20.** According to the collision theory and kinetic theory, increase in temperature does the following except

**A.** Increases the frequency of collision

**B.** Increases the number of particles with energy greater than the activation energy

**C.** Increases the activation energy

**D.** Increases the kinetic energy of the molecules

**21.** The part of the universe that is of immediate interest in a reaction is termed

**A.** Universe

**B.** Surrounding

**C.** System

**D.** Environment

**22.** An example of a natural radioactivity is

**A.** ${}^{14}_{7}N + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$

**B.** ${}^{214}_{83}Bi \rightarrow {}^{214}_{84}Po + {}^{0}_{-1}e$

**C.** ${}^{238}_{92}U + {}^{4}_{2}He \rightarrow {}^{17}_{8}O + {}^{1}_{1}H$

**D.** ${}^{12}_{6}C \rightarrow {}^{12}_{7}N + {}^{1}_{-1}e$

**23.** The general gas equation can be used to calculate the molar mass, $M_r$ value of a gas. For a sample of a gas of mass mg, which expression will give the value of $M_r$?

**A.** $M_r = \frac{PVRT}{m}$

**B.** $M_r = \frac{mPV}{RT}$

**C.** $M_r = \frac{mRT}{PV}$

**D.** $M_r = \frac{PV}{mRT}$

**24.** Which of these bonds does not exist within a molecule of ammonium chloride?

**A.** Ionic

**B.** Covalent

**C.** Dative covalent

**D.** Van der waals

**25.** The dissolution of a non-volatile solute in a volatile solvent

**A.** Increases its boiling point

**B.** Decreases its boiling point

**C.** Has no effect

**D.** Increases the vapour pressure

**26.** Gaseous deviate from Charles' law at

**A.** High temperature

**B.** Low pressure

**C.** Low temperature

**D.** Moderate pressure

**27.** The average distance of the orbital from the nucleus of an atom orbital is determined by the

**A.** Spin quantum number

**B.** Principal quantum number

**C.** Magnetic quantum

**D.** Orbital subshell

**28.** What is the standard cell potential for the overall reaction of the cell below?

$Cu(s)/Cu^{2+}\ (1M)//Ag^{+}\ (1M)/Ag(s)$

$[E^{0}_{redox}Cu^{2+}\ (aq)/Cu(s)\ =\ 0.3402V\ and\ E^{0}Ag^{+}(aq)/Ag(s)\ =\ 0.7996V]$

**A.** $0.229V$

**B.** $1.2590V$

**C.** $0.4594V$

**D.** $1.1398V$

**29.** The element that gives no colour in flame is

**A.** $Be$

**B.** $Ca$

**C.** $Na$

**D.** $Li$

**30.** All group I metals react with water to form

**A.** Hydrogen gas only

**B.** Neutral solutions

**C.** Acidic solution with liberation of hydrogen

**D.** Strong caustic solution with liberation of hydrogen

**31.** The effect of acid rain includes the following except

**A.** Death of aquatic animals

**B.** Damage of buildings and status

**C.** Corrosion of iron

**D.** Depletion of ozone layer

**32.** Which of the following is not a greenhouse gas?

**A.** $CFC_S$

**B.** $SO_2$

**C.** $H_2O$

**D.** $CH_4$

**33.** Which of the following is the correct increasing order of conductivity in aqueous solution?

**A.** $Li^+, Na^+, K^+, Cs^+, Rb^+$

**B.** $Rb^+, Na^+, K^+, Cs^+, Li^+$

**C.** $Li^+, Na^+, K^+, Rb^+, Cs^+$

**D.** $Na^+, K^+, Cs^+, Li^+, Rb^+$

**34.** The alkaline earth metals become more reactive as we

**A.** Ascend the group

**B.** Descend the group

**C.** Introduce other group elements

**D.** Introduce an external force

**35.** Chromium has the following isotopic masses and fractional abundances

| $Mass\ Number$ | $Isotopic\ Mass\ (amu)$ | $Fractional\ Abundance$ |
|---|---|---|
| 50 | 49.9461 | 0.0435 |
| 52 | 51.9405 | 0.8379 |
| 53 | 52.9407 | 0.0950 |
| 54 | 53.9389 | 0.0236 |

What is the atomic mass of chromium?

**A.** 55.49 $amu$

**B.** 50.49 $amu$

**C.** 51.99 $amu$

**D.** 51.00 $amu$

**36.** What is the name given to this complex ion $[Co(SO_4)(NH_3)_5]^+$?

**A.** Pentaaminesulphatocobalt(I)ion

**B.** Pentaaminesulphatocobaltate(II)ion

**C.** Pentaaminesulphatocobalt(III)ion

**D.** Pentaaminesulphatocobalt(IV)ion

**37.** Which of the following will form pale yellow precipitate slightly soluble in aqueous ammonia upon treatment with $trioxonitrate\ (V)$ acid followed by silver nitrate

**A.** $NAHCO_3$

**B.** $NaCI$

**C.** $NaBr$

**D.** $Nal$

**38.** Which of the statement below is not a property of halogen?

**A.** They all exhibit valency of $+1$

**B.** They belong to the $P - Block$ of the periodic table

**C.** They are non-metals

**D.** They are all oxidizing agents

**39.** An element which can be detected using Lassaigne's test is

**A.** Sodium

**B.** Sulphur

**C.** Potassium

**D.** Magnesium

**40.** If the molecule A-B breaks homolytically, then

**A.** A becomes negatively charged

**B.** A becomes positively charged

**C.** Electrons are shared evenly

**D.** Both electrons move to B because B is more electronegative

**41.** Lucas reagent is used to distinguish between

**A.** Saturated and unsaturated hydrocarbons

**B.** Primary and secondary amines

**C.** Alkanals and alkanones

**D.** The three classes of alkanols

**42.** The IUPAC name of the compound $CH = C - CH_2 - CH_2 - COOH$ is

**A.** $1 - pentyne - 4 - oic\ acid$

**B.** $Pentyne - 1 - oic\ acid$

**C.** $5 - pentyn - 1 - oic\ acid$

**D.** $Pent - 4 - yne - 1 - oic\ acid$

**43.** Which of the following compounds will show geometrical isomerism?

**A.** $1 - butane$

**B.** $2 - butene$

**C.** $2 - pentyne$

**D.** $1,2 - dichloropropane$

**44.** The number of $\pi$ electrons in butyne is

**A.** 6

**B.** 2

**C.** 4

**D.** 8

**45.** The change in hybridization of carbon when two moles of hydrogen gas is passed over ethyne in the presence of platinum is

**A.** $Sp^2\ to\ Sp^3$

**B.** $Sp\ to\ Sp^3$

**C.** $Sp^3\ to\ Sp$

**D.** $Sp\ to\ Sp^2$

**46.** $2.40g$ of ethanol was reacted with an excess of aqueous acidified $potassium\ dichromate\ (VI)$. The yield of the desired organic product was $70.0\%$. What mass of the desired product was collected?

$[C = 12, O = 16, H = 1]$

**A.** $1.60g$

**B.** $4.10g$

**C.** $2.29g$

**D.** $3.29g$

**47.** In the reaction between propene and hydrogen chloride, the electrophile that initiates the reaction is

**A.** The $\pi$ electrons of $C = C$ bond in propane

**B.** The partially positive hydrogen atom from HCI

**C.** The partially negative chlorine atom from HCI

**D.** The fully positive hydrogen from HCI

**48.** $Iodohexane\ and\ 2-iodooctane$ are related as

**A.** Functional isomers

**B.** Positional isomers

**C.** Chain isomers

**D.** Homologues

**49.** Which of the following has the shortest carbon-carbon bond length?

**A.** Propene

**B.** Propane

**C.** Propyne

**D.** Ethane

**50.** The properties of organic molecules are related to Carbon to Carbon bond properties such as:

(I) Bond Length

(II) Bond dissociation energy

(III) Bond rotation

Which of these properties reveal the fact that a $\pi - \ bond$ is weaker than a $\sigma - bond$?

**A.** (II) Only

**B.** (III) Only

**C.** (I) Only

**D.** (I) and (III)

**SECTION B: ESSAY QUESTIONS**

**Answer FOUR Questions in all; Not more than ONE Question from each course**

**CHM 001: GENERAL CHEMISTRY**

**1.a** What are acid-base indicators? $(1\ mark)$

**1.b** A $550mg$ tablet containing vitamin C, $C_6H_8O_6$, was dissolved in water and reacted with bromine ($C_6H_8O_6,\ + Br_2\ \rightarrow\ C_6H_6O_6\ +\ 2HBr$). The resulting solution was titrated with $40cm^3\ of\ 0.145M\ NaOH$

$(HBr\ +\ NaOH\ \rightarrow\ NaBr\ +\ H_2O)$

$[C\ =\ 12, Na\ =\ 23, O\ =\ 16, H\ =\ 1, Br\ =\ 79]$

**i)** Calculate the percentage of vitamin C in the tablet. Calculate the percentage of Carbon in the Vitamin C content of the tablet.

$(6\ marks)$

**1.c** The equilibrium-constant expression for a reaction is

$K_C\ = \frac{[NO_2]^4[O_2]}{[N_2O_5]^2}$

**i)** Write the balanced chemical equation corresponding to this expression.

**ii)** What is the equilibrium constant expression when the concentrations are halved?

**iii)** Write the equilibrium constant expression for c (ii) above when the equation is reversed

$(3\ marks)$

**2.a**

**i)** Define an acid and a base in terms of Browsted-Lowry concept

$(2\ marks)$

**ii)** Calculate the pH of a buffer solution containing $0.623moldm^{-3}\ hexanoic\ acid\ and\ 0.812moldm^{-3}\ sodium\ hexanoate.$ $[Ka\ hexanoic\ acid\ =\ 1.32\ x\ 10-5].$ $(2\ marks)$

**2.b** Classify the reactants in each of the following reactions as acid or base and identify the acid-base conjugate pairs:

**i)** $NH_4^+ + H_2O \rightleftharpoons NH_3 + H_2O^+$ $(2\ marks)$

**ii)** $OH^- + CH_3NH_2 \rightleftharpoons CH_3NH_3^+ + O^{2-}$ $(2\ marks)$

**2.c** Using relevant reactions equations, explain why

**i)** Water is amphiprotic $(1\ mark)$

**ii)** Methanol is a Bronsted-Lowry acid $(1\ mark)$

**CHM 002: PHYSICAL CHEMISTRY**

**3.a** Using the following data

$Fe^{3+}_{(aq)} + e^- \rightleftharpoons Fe^{2+}_{(aq)}; \quad E^0 = +0.77V$

$MnO_{4(aq)} + 8H^+_{(aq)} + 5e^- \rightleftharpoons Mn^{2+}_{(aq)} + 4H_2O_{(l)}; \quad E^0 = +1.52V$

$I_{2(aq)} + 2e^- \rightleftharpoons 2I^-_{(aq)}; \quad E^0 = +0.54V$

$Cr_2O^{2-}_{7(aq)} + 14H^+_{(aq)} + 6e^- \rightleftharpoons 2Cr^{5+} + 7H_2O_{(l)}; \quad E^0 = +1.33V$

Predict the outcome of mixing aqueous solutions of

**i)** Iron (II) sulphate with acidified potassium manganese (VII) $(1\ mark)$

**ii)** Potassium iodide with acidified potassium dichromate (VI) $(1\ mark)$

**3.b** Write the half equations and the balanced equations for the reactions you predict above. $(2\ marks)$

**3.c** Calculate the standard cell potential in each case $(2\ marks)$

**3.d** The reaction $\frac{1}{2}N_2 + \frac{3}{2}H_2 \rightarrow NH_2$ proceeded in a closed autoclave of volume $5dm^{-3}\ in\ 1\ second$, $0.01\ mole$ of nitrogen reacted. Calculate the rate of

**i)** Consumption of nitrogen $(1\ mark)$

**ii)** Consumption of hydrogen $(1\ mark)$

**iii)** Formation of ammonia $(1\ mark)$

**iv)** The reaction $(1\ mark)$

**4.a** State two differences between alpha (α) and beta ($\beta$) particles.

$(2\ marks)$

**4.b**

**i)** Define nuclear fusion

**ii)** Calculate the nuclear energy per mole for the following nuclear reaction:

$^{14}_{7}N + {}^{4}_{2}He \rightleftharpoons {}^{17}_{8}O + {}^{1}_{1}H$

Given the following isotropic masses

$^{14}_{7}N = 13.9992amu$

$^{4}_{2}He = 4.0026amu$

$^{17}_{8}O = 16.9991amu$

$^{1}_{1}H = 1.0078amu$

$1amu = 1.66 \times 10^{-27} kg\ atom^{-1}$

$N_A = 6.023 \times 10^{23}\ atom\ mol^{-1}$

**4.c**

**i)** State Hess's Law

**ii)** For the reaction:

$C_2H_{4(g)} + 3O_{2(g)} \rightarrow 2CO_{2(g)} + 2H_2O_{(g)}$

$\Delta H^0_{rxn} = -1323kJmol^{-1}$

$Given\ that\ S^0[C_2H_{4(g)}] = 219.6\ \text{Jmol}^{-1}$;

$S^0[O_{2(g)}] = 205\ \text{Jmol}^{-1}$

$S^0[CO_{2(g)}] = 213.6\ \text{Jmol}^{-1}$

$S^0[H_2O_{(g)}] = 130.6\ \text{Jmol}^{-1}$, Calculate the value of $\Delta G^0_{rxn}\ at\ 250$

$(3\ marks)$

**CHM 003: INORGANIC CHEMISTRY**

**5.a** The elements in Group II show a variety of trends in their properties

**i)** Explain the trend in reactivity of magnesium, calcium and barium with cold water.

$(2\ marks)$

**ii)** Write the equation for the reaction of magnesium with steam

$(1\ mark)$

**5.b** Name an element among the period three elements which forms an amphoteric oxide and write equations which illustrate this amphoteric property.

$(2\ marks)$

**5.c**

**i)** What is meant by the term ionization energy?

$(1\ mark)$

**ii)** Explain why the first ionization energy of nitrogen is greater than that of oxygen.

$(2\ marks)$

**5.d** List four unique properties of carbon that makes it different from other members of group 14 elements. $(2\ marks)$

**6.a**

**i)** What is a Complex ion? $(1\ mark)$

**ii)** What is the name given to the molecules of ions bonded to a complex ion?

$(1\ mark)$

**6.b**

**i)** What is the co-ordination number of the metal in the following complexes

$[Cu(NH_3)_4(H_2O)_2]^{2+}$ and

$[Cu(H_2NCH_2CH_2NH_2)_2]^{2+}$ $(1\ mark)$

**ii)** Provide the IUPAC names for the following:

**(1)** $[Fe(H_2O)_2(en)_2]^{2+}$

**(2)** $[Ni(NH_3)_2Cl_4]^{2-}$

**(3)** $[Cr(NH_3)_3(H_2O)_3]Cl_3$

**(4)** $[Co(H_2NCH_2CH_2NH_2)_3(SO_4)_3]$ $(2\ marks)$

**6.c**

**i)** Briefly explain why beryllium tends to be different from the elements in group 2 of the periodic table. Give two anomalous properties of beryllium $(2\ marks)$

**ii)** Explain the trend of the thermal decomposition of the carbonates and nitrates of the alkaline earth metals. $(2\ marks)$

## CHM 004: ORGANIC CHEMISTRY

**7.a**

**i)** What is isomerism? $(1\ mark)$

**ii)** Draw the structural formula of the isomeric alcohols with the molecular formula $C_4H_{10}O$ and name them using the IUPAC system of naming. $(4\ marks)$

**iii)** What are the products of oxidation of each of the isomers in $1a\ (ii)$?

$(2\ marks)$

**7.b** Show the conversion of ethanoic acid into the following with suitable equation and reagents;

**i)** Acid chloride (II) Acid anhydride (III) Ester $(3\ marks)$

**8.a**

**i)** What is a Grignard reagent?

$(1\ mark)$

**ii)** Give an equation for the preparation of Grignard reagent.

$(1\ mark)$

**iii)** What is the product of the reaction?

$$CH_3CH_2CHO\ + CH_3CH_2MgBr\ \rightarrow ?\ + ?$$

Give the IUPAC name of the product

$(1\ mark)$

**8.b** Reactions of alcohols involve cleavage of the $C - O$ bond or that of the $O - H$ bond. Give one example each of such reactions

$(2\ marks)$

**8.c** Why does benzene not undergo addition reaction as expected of unsaturated hydrocarbons? $(1\ mark)$

**8.d** State two (2) reasons why aldehydes are more reactive than ketones.

$(2\ marks)$

**8.e** Name the monomers used in the synthesis of $Nylon\ 6{,}6$. $(1\ mark)$

**SECTION A: MULTIPLE CHOICE ANSWERS**

**1.** Option C

**2.** Option D

**3.** Option A

**4.** Option B

**5.** Option B

**6.** Option C

$[OH^-] = 1.5 \times 10^{-9}\, moldm^{-3}$

$P^{OH} = -\log[OH^-]$

$= -\log[1.5 \times 10^{-9}]$

$P^{OH} = 8.82$

$\therefore Since\ P^{OH}\ is\ greater\ than\ 7\ (neutral), the\ solution\ is\ basic$

**7.** Option A

**8.** Option C

$Using\ degeneracy\ of\ orbitals = 2l + 1$

$Where\ l = Angular\ momentum\ quantum\ number = 2$

$Degeneracy\ of\ orbitals = 2 \times 2 + 1 = 4 + 1 = 5$

**9.** Option C

**10.** Option A

$Let\ the\ O.N\ of\ O\ be\ x$

$O.N\ of\ F = -1$

$x + (-1) \times 2 = 0$

$x - 2 = 0$

$x = +2$

**11.** Option D

**12.** Option D

**13.** Option D

**14.** Option C

**15.** Option B

$Na_{(s)} + H_2O_{(l)} \rightarrow NaO_{(s)} + H_{2(g)}$

$Volume\ of\ H_{2(g)}\ liberated = 500cm^3$

$Recall, no\ of\ mole = \dfrac{Volume\ of\ gas}{Molar\ volume\ of\ gases}$

$no\ of\ mole\ H_{2(g)} = \dfrac{500}{22400} = 0.223mol$

$From\ the\ equation\ of\ reaction:$

$mole\ ratio: \underbrace{Na}_{1} : \underbrace{H_2}_{1}$

$\therefore no\ of\ mole\ of\ Na_{(s)} = 0.223\text{mol}$

$no\ of\ mole = \dfrac{mass}{molar\ mass}$

$mass\ of\ Na = no\ of\ mole \times R.A.M$

$= 0.223 \times 23$

$= 0.51g$

**16.** Option A

$Mass\ of\ water\ of\ Crystallization, 7H_2O$

$= 7(2 \times 1 + 16) = 7(2 + 16) = 7 \times 18$

$= 126g$

$Mass\ of\ hydrated, MgSO_4.7H_2O$

$= 24 + 32 + 64 + 126 = 246g$

$Percentage\ by\ Mass\ of\ water\ of\ Crystallization = \dfrac{126}{246} \times 100 = 51.2\%$

**17.** Option A

$C_1 = 5moldm^{-3}, C_2 = 0.5moldm^{-3},$

$V_2 = 300cm^3, V_1 = ?$

$Using\ C_1V_1 = C_2V_2$

$5V_1 = 0.5 \times 300$

$5V_1 = 150$

$V_1 = \frac{150}{5} = 30cm^3$

**18.** Option A

$Mass\ of\ iron\ complex$

$= 12 \times 34 + 1 \times 32 + 56 + 4 \times 14 + 4 \times 16$

$= 408 + 32 + 56 + 56 + 64 = 616g$

$\%\ by\ mass\ of\ Fe\ in\ the\ complex$

$= \frac{56}{616} \times 100 = 9.1\%$

$Mass\ of\ haemoglobin = 0.75g$

$Mass\ of\ complex\ in\ the\ haemoglobin$

$= \frac{6}{100} \times 0.75 = 0.045g$

$Mass\ of\ Fe\ contributed\ by\ the\ complex$

$= \frac{9.1}{100} \times 0.045 = 0.0041g$

**19.** Option A

**20.** Option C

**21.** Option C

**22.** Option B

**23.** Option C

$The\ general\ gas\ equation\ is\ given\ by; PV = nRT$

$But\ n = \frac{m}{M_r}$

$PV = \frac{m}{M_r}RT$

$M_rPV = mRT$

$M_r = \frac{mRT}{PV}$

**24.** Option D

**25.** Option A

**26.** Option C

**27.** Option D

**28.** Option D

$Reduction\ reaction$:

$Cu^{2+}_{(aq)} + 2e^- \longrightarrow Cu_{(s)} \quad E^{\theta}_{red} = +0.3402V$

$Oxidation\ reaction$:

$Ag_{(s)} \longrightarrow Ag^{+}_{(aq)} + e^- \quad E^{\theta}_{ox} = -0.7996V$

$Overall\ reaction$:

$2Ag_{(s)} + Cu^{2+} \longrightarrow 2Ag + + Cu_{(s)}$

$E^{\theta}_{cell} = E^{\theta}_{red} - E^{\theta}_{ox}$

$E^{\theta}_{cell} = +0.3402 - (-0.7996)$

$= +0.3402 + 0.7996$

$= +1.1398V$

**29.** Option A

**30.** Option D

**31.** Option D

**32.** Option B

**33.** Option C

**34.** Option A

**35.** Option C

$Let\ the\ isotopic\ mass\ and\ fractional\ abundance\ be\ x\ and\ y\ respectively.$

$Average\ atomic\ mass$

$$= \sum_{i=1}^{n} (Isotopic\ mass)_i (Fractional\ abundance)_i$$

$$= \sum_{i=1}^{4} X_i Y_i$$

$$= \sum_{i=1}^{4} = (x_i y_i) + (x_2 y_2) + (x_3 y_3) + (x_4 y_4)$$

$$= \sum_{i=1}^{4} = (49.9461 \times 0.0435) + (51.9405 \times 0.8379) + (52.9407 \times 0.0950) + (53.9389 \times 0.0236)$$

$$= \sum_{i=1}^{4} = 2.17 + 43.52 + 5.03 + 1.27 = 51.99amu$$

**36.** Option A

Let the O.N of Cobalt be $x$

$x + (-2) = 1$

$x - 2 = 1$

$x = +3$

Cobalt (I) ion

**37.** Option D

**38.** Option A

**39.** Option B

**40.** Option C

**41.** Option D

**42.** Option B

**43.** Option D

$The\ two\ geometric\ isomers\ Trans\ and\ Cis\ are$:

$$\begin{array}{l} \quad H \quad Cl \quad H \\ \quad | \quad | \quad | \\ H-C-C-C-H \\ \quad | \quad | \quad | \\ \quad Cl \quad H \quad H \end{array}$$

Trans -1, 2 – dichloropropane

$$\begin{array}{l} \quad Cl \quad Cl \quad H \\ \quad | \quad | \quad | \\ H-C-C-C-H \\ \quad | \quad | \quad | \\ \quad H \quad H \quad H \end{array}$$

C is -1, 2 – dichloropropane

**44.** Option C

$$\begin{array}{l} \quad H \qquad\qquad H \\ \quad | \qquad\qquad | \\ H-\overset{\pi}{\underset{\pi}{C}}-C \equiv C-C-H \\ \quad | \qquad\qquad | \\ \quad H \qquad\qquad H \end{array}$$

No of pie $(\pi)$ bond in $butyne = 2$

Each $\pi - bond$ have 2 electrons

The no of $\pi$ electrons = $2 \times 2 = 4$

**45.** Option D

$C_2H_2 + H_{2(g)} \rightarrow C_2H_{4(g)}$

$$H-C-C-H + H_{2(g)} \overset{Pt}{\rightarrow} \underbrace{\begin{array}{l} \quad\quad\quad H \\ \quad\quad\quad | \\ H=C-C-H \\ \quad | \quad | \\ \quad H \quad H \end{array}}_{Ethene\ Hybridized}$$

$\underbrace{Ethyne}_{Sp} \rightarrow \underbrace{Ethene}_{Sp^2}$

**46.** Option C

$$\underbrace{C_2H_5OH}_{ethanol} \overset{K_2Cr_2O_7/2H^+}{\rightleftarrows} \underbrace{CH_3COOH}_{ethanoic\ acid}$$

$Complete\ oxidation\ of\ a\ primary\ alchohol$

$mass\ of\ C_2H_5OH = 2.40g$

$molar\ mass = (2 \times 12) + (5 \times 1) + 16 + 1$

$= 46gmol^{-1}$

$mass\ of\ product = no\ of\ mole \times molar\ mass$

$= 0.0522 \times 60$

$= 3.132g$

$\therefore Since\ 70\%\ of\ the\ desired\ product\ was\ yieled$

$mass\ of\ product\ yielded = \frac{70}{100} \times 3.132$

$= 2.19g$

**47.** Option B

**48.** Option D

**49.** Option C

| $Bond\ Type$ | $Covalent\ bond\ length\ (PM)$ |
|---|---|
| $C - C$ | 154 |
| $C = C$ | 133 |
| $C \equiv C$ | 120 |

**50.** Option A

The $C - C$ sigma bond has bond dissociation energy of $88Kcal/mole$ while the sigma bond of $C = C$ double bond has bond dissociation energy of $95Kcal/mole$.

Therefore, more overlap leads to stronger bonds.

**SECTION B: ESSAY ANSWERS**

**CHM 001: GENERAL CHEMISTRY**

**1.a** Acid – Base indicators are Substance usually weak acids or bases which when dissolves in water dissociate slightly, forms ions and change colour with pH.

**1.b**

**i)** $Mass\ of\ tablet\ =\ 550mg\ =\ 0.55g$

$Molar\ mass\ of\ Vitamin\ C\ (C_6H_8O_6)$

$=\ 6\ \times\ 12\ +\ 8\ \times\ 1\ +\ 6\ \times\ 16\ =\ 72\ +\ 8\ +\ 96$

$=\ 176gmol^{-1}$

$Molarity\ of\ NaOH\ =\ 0.145M\ (0.145moldm^{-3})$

$Volume\ of\ NaOH\ =\ 40cm^3\ =\ 40\ \times\ \frac{1}{1000}\ dm^3$

$=\ 0.004\ dm^3$

$Molarity\ of\ NaOH\ =\ \frac{No\ of\ mole}{Volume}$

$No\ of\ mole\ =\ Molarity\ \times\ Volume$

$=\ 0.145\times\ 0.004$

$No\ of\ mole\ =\ 0.0058mole$

$From\ equation\ of\ reaction,$

$HBr\ +\ NaOH\ \longrightarrow NaBr\ +\ H_2O$

$1\ mole\ of\ NaOH\ requires\ 1\ mole\ of\ HBr$

$0.0058mole\ of\ NaOH\ will\ requires\ 0.0058mole\ of\ HBr$

$From\ equation\ of\ reaction,$

$C_6H_8O_6\ +\ Br_2\longrightarrow C_6H_6O_6\ +\ 2HBr$

$2mole\ of\ HBr\ was\ produced\ by\ 176g\ of\ C_6H_8O_6$

$0.0058 mole\ of\ HBr\ will\ be\ produced\ by$

$$\frac{0.0058 \times 176}{2} = 0.51g\ of\ Vitamin\ C$$

$\%\ by\ mass\ of\ Vitamin\ C\ in\ the\ tablet$

$$= \frac{0.51}{0.55} \times 100 = 92.7\%$$

**ii)** $Mass\ of\ carbon\ in\ Vitamin$

$C = 6 \times 12 = 72g$

$\%\ by\ mass\ of\ carbon\ in\ the\ vitamin\ C$

$$= \frac{72}{176} \times 100 = 40.9\%$$

**1.c**

**i)** $2N_2O_5 \rightleftharpoons 4NO_2 + O_2$

**ii)** $When\ the\ concentration\ are\ halved$

$$N_2O_5 \rightleftharpoons 2NO_2 + \frac{1}{2}O_2$$

$$K_c = \frac{[NO_2]^2[O_2]^{\frac{1}{2}}}{[N_2O_5]}$$

**iii)** $When\ the\ equation\ is\ reversed,$

$$2NO_2 + \frac{1}{2}O_2 \rightleftharpoons N_2O_5$$

$$K_c = \frac{[N_2O_5]}{[NO_2]^2[O_2]^{\frac{1}{2}}}$$

**2.a**

**i)** An acid can be defined as a substance that donates a proton while base can be defined as a substance that accepts a proton in a reaction

**ii)** Concentration of the acid

$HA = 0.623 moldm^{-3}$

Concentration of the Salt,

$A^- = 0.812 moldm^{-3}$

$K_a = 1.32 \times 10^{-5}$

$pK_a = -log\ K_a$

$pK_a = -log\ (1.32 \times 10 - 5)$

$pK_a = -(-4.9) = 4.9$

$Using\ Buffer\ equation$

$pH = PK_a + \log\frac{[A^-]}{[HA]}$

$pH = 4.9 + \log\frac{0.812}{0.623}$

$pH = 4.9 + log\ (1.3034)$

$pH = 4.9 + 0.12 = 5.04$

$\therefore pH\ of\ the\ solution = 5.04$

**2.b**

**i)**

| $Acid$ | $Base$ |
|---|---|
| $NH_4^+$ | $NH_3$ |

In the reaction, the two conjugates acid/base pairs are $NH_4^+/NH_3$ and $H_3O^+/H_2O$

**ii)**

| Acid | Base |
|---|---|
| $CH_3NH_3^+$ | $CH_3NH_2$ |

The two-conjugate acid/base pairs are

$CH_3NH_2/CH_3NH_3^+ \; and \; OH^-/O^{2-}$

**2.c**

**i)** Water is amphiprotic because it can act as either a Bronsted – Lowry acid or Bronsted – Lowry base. In other word, water can donate as well can accept proton.

$H_2O_{(l)} + NH_{3(aq)} \rightleftharpoons OH^-_{(aq)} + NH^+_{4(aq)}$

$H_2O_{(l)} + HCl_{(aq)} \rightleftharpoons H_3O^+_{(aq)} + Cl^-_{(aq)}$

**ii)** Methanol is a Bronsted – Lowry acid because is a proton donor. In other words, it fairly easily donates $O-H$ proton to strong bases.

$CH_3OH + NaOH \rightleftharpoons NaOCH_3 + H_2O$

$CH_3O^-_{(aq)} + H^+_{(aq)} + Na^+_{(aq)} + OH^-_{(aq)} \rightleftharpoons NaOCH_{3(aq)} + H_2O_{(l)}$

**CHM 002: PHYSICAL CHEMISTRY**

**3.a**

**i)** $Fe^{2+}_{(aq)} + H_{(aq)} + MnO^{-}_{4(aq)} \rightarrow Fe^{3+}_{(aq)} + Mn^{2+}_{(aq)} + H_2O_{(l)}$

**ii)** $I^{-}_{(aq)} + H^{+}_{(aq)} + Cr_2O^{2-}_{7(aq)} \longrightarrow I_{2(aq)} + Cr^{3+}_{(aq)} + H_2O_{(l)}$

**3.b**

**i)** The two half equations for the reduction reaction (i) and (ii) for the oxidation reaction

$\underbrace{MnO^{-}_{4(aq)}}_{+7} \longrightarrow \underbrace{Mn^{2+}}_{+2} \ (Reduction)$

$MnO^{-}_{4(aq)} \longrightarrow Mn^{2+} + H_2O$

$MnO^{-}_{4(aq)} + 8H^{+} \longrightarrow Mn^{2+} + 4H_2O$

$MnO^{-}_{4(aq)} + 8H^{+} + 5e^{-} \longrightarrow Mn^{2+} + 4H_2O \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ (i)$

$\underbrace{Fe^{2+}_{(aq)}}_{+2} \longrightarrow \underbrace{Fe^{3+}_{(aq)}}_{+3} \ (Oxidation)$

$Fe^{2+}_{(aq)} \longrightarrow Fe^{3+}_{(aq)} + e^{-}$

$5Fe^{2+}_{(aq)} \longrightarrow 5Fe^{3+}_{(aq)} + 5e^{-} \ \ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ldots\ (ii)$

$Adding\ eqn\ (i)\ and\ (ii)$

$MnO^{-}_{4(aq)} + 8H^{+} + 5e^{-} \longrightarrow Mn^{2+} + 4H_2O$

$5Fe^{2+}_{(aq)} \longrightarrow 5Fe^{3+}_{(aq)} + 5e^{-}$

$Overall\ equation$:

$5Fe^{2+}_{(aq)} + MnO^{-}_{4(aq)} + 8H^{+} \longrightarrow 5Fe^{3+}_{(aq)} + Mn^{2+} + 4H_2O$

**ii)** The two half equations (iii) for the reduction reaction and (iv) for oxidation reaction

$\underbrace{Cr_2O^{2-}_{7}}_{+7} \longrightarrow \underbrace{2Cr^{3+}}_{+3} \ (Reduction)$

$Cr_2O^{2-}_{7} \longrightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 8H^+ \longrightarrow 2Cr^{3+} + 7H_2O$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$ …… $(iii)$

$2I^-_{(aq)} \longrightarrow I_{2(aq)}\ (Oxidation)$

$2I^- \longrightarrow I_2 + 2e^-$

$6I^- \longrightarrow 3\,I_2 + 6e^-$ …… $(iv)$

$Adding\ (iii)\ and\ (iv)$

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

$6I^- \longrightarrow 3I_2 + 6e^-$

$Overall\ equation$:

$6I^- + Cr_2O_7^{2-} + 14H^+ \longrightarrow 3I_2 + 2Cr^{3+} + 7H_2O$

**3.c**

**i)** Reduction

$MnO_4^- + 8H^+ + 5e^- \longrightarrow Mn^{2+} + 4H_2O$

$E^\theta_{red} = +1.52V$

Oxidation

$Fe^{2+}_{(aq)} \longrightarrow Fe^{3+}_{(aq)} + e^-$

$E^\theta_{ox} = -E^\theta_{red} = -0.77V$

Standard cell potential

$E^\theta_{cell} = E^\theta_{red} + E^\theta_{ox}$

$E^\theta_{cell} = +1.52 - 0.77 = +0.75V$

**ii)** Reduction

$Cr_2O_7^{2-} + 14H^+ + 6e^- \longrightarrow 2Cr^{3+} + 7H_2O$

$E^\theta_{red} = +1.33V$

Oxidation

$2I^- \longrightarrow I_2 + 2e^-$

$E^{\theta}_{ox} = -E^{\theta}_{red} = -0.54V$

Standard cell potential

$E^{\theta}_{cell} = E^{\theta}_{red} + E^{\theta}_{ox}$

$E^{\theta}_{cell} = +1.33 - 0.54 = +0.79V$

**3.d** $Volume\ of\ nitrogen = 5dm^3$

$No\ of\ mole\ of\ nitrogen = 0.01mole$

$Concentration\ of\ nitrogen,$

$\Delta[N_2] = \frac{No\ of\ mole}{Volume} = \frac{0.01}{5}$

$\Delta[N_2] = 0.002moldm^{-3}$

**i)** The rate of consumption of nitrogen

$= \frac{\Delta[N_2]}{\Delta t} = \frac{0.002}{1} = 0.002moldm^{-3}s^{-1}$

**ii)** From the stoichiometry equation,

$If\ \frac{1}{2}\ mole\ of\ N_2\ consumes\ \frac{3}{2}\ moles\ of\ H_2$

$0.002mole\ of\ N_2\ will\ consumed\ \frac{0.002 \times \frac{3}{2}}{\frac{1}{2}} = 0.002 \times 3 = 0.006mole$

$The\ rate\ of\ consumption\ of\ hydrogen = 0.006moldm^{-3}s^{-1}$

**iii)** $If\ \frac{1}{2}\ mole\ of\ N_2\ forms\ 1\ mole\ of\ NH_3$

$0.002mole\ of\ N_2\ will\ forms\ \frac{0.002 \times 1}{\frac{1}{2}}$

$= 0.002 \times 2 = 0.004 mole$

$The\ rate\ of\ formation\ of\ ammonia$

$= 0.004 moldm^{-3}s^{-1}$

**iv)** The rate of reaction,

$$r = \frac{Concentration}{Time}$$

$$r = \frac{-\Delta[N_2]}{\frac{1}{2}\Delta t} = \frac{-\Delta[H_2]}{\frac{1}{2}\Delta t} = \frac{\Delta[N_3]}{\Delta t}$$

$$r = \frac{0.002}{\frac{1}{2}} = \frac{0.006}{\frac{3}{2}} = \frac{0.004}{1} = 0.004 moldm^{-3}s^{-1}$$

**4.a**

| $Alpha\ (\alpha)\ Particles$ | $Beta\ (\beta)\ Particles$ |
|---|---|
| It consists of several nucleons | It consists of only one nucleon |
| It has a relatively low penetrating power | It has a medium penetrating power |
| There is only one type of alpha particle (alpha decay) | There are two types of beta particles ($electron,$ $\beta^-\ decay\ and\ positron\ decay, \beta^+$ ) |

**4.b**

**i)** Nuclear fusion is a reaction in which two or more atomic nuclei are combined or fused to form one heavier nucleus with the release of energy.

**ii)** Mass of reactants

$m_R = 13.9992 + 4.0026$

$= 18.0018amu$

Mass of Products

$m_P = 16.9991 + 1.0078$

$= 18.0069amu$

$Mass\ difference, m = m_P - m_R$

$m = 0.0051amu$

$m = 0.0051 \times 1.66 \times 10^{-27} Kgatom^{-1}$

$m = 8.466 \times 10^{-30} Kgatom^{-1}$

Using Einstein relation

$Nuclear\ energy, E = mc^2$

Where m = mass difference,

$C = speed\ of\ light = 3 \times 10^8 ms^{-1}$

$E = 8.466 \times 10^{-30} \times (3 \times 108)^2$

$E = 8.466 \times 10^{-30} \times 9 \times 10^{16}$

$E = 7.6194 \times 10^{-13} Jatom^{-1}$

$Nuclear\ energy\ per\ mole$

$7.6194 \times 10^{-13} Jatom^{-1} \times 6.023 \times 10^{23} atomsmol^{-1}$

$= 4.5892 \times 1011 Jmol^{-1}$

**4.c**

**i)** Hess's law states that regardless of the multiple stages or steps of a reaction, the total enthalpy change for the reaction is the sum of all changes

**ii)** Applying Hess's law

$\Delta G^{\theta}_{rxn} = \Delta H^{\theta}_{rxn} - T\Delta S^{\theta}_{rxn}$

$Where\ \Delta H^{\theta}_{rxn} = -1323 KJmol^{-1}$

$\Delta H^{\theta}_{rxn} = -1323000 Jmol^{-1}$

$T = (\theta + 273)K$

$T = 25 + 273 = 298K$

$\Delta S^{\theta}_{rxn} = \sum nS^{\circ}_{p} - \sum mS^{\circ}_{R}$

$\Delta S^{\theta}_{rxn} = [2S^{\circ}[CO_{2(g)}] + 2S^{\circ}[H_2O_{(g)}] - [S^{\circ}[C_2H_{4(g)}] + 3S^{\circ}[O_{2(g)}]]$

$\Delta S^{\theta}_{rxn} = [2 \times 213.6 + 2 \times 130.6] - [219.6 + 3 \times 205]$

$\Delta S^{\theta}_{rxn} = [427.2 + 216.2] - [219.6 + 615]$

$\Delta S^{\theta}_{rxn} = 643.4 - 834.6$

$\therefore \Delta S^{\theta}_{rxn} = -191.2 Jmol^{-1}$

$\Delta G^{\theta}_{rxn} = -132300 - 298(-191.2)$

$\Delta G^{\theta}_{rxn} = -132300 + 569776$

$\Delta G^{\theta}_{rxn} = 437476 Jmol^{-1}$

$\therefore \Delta G^{\theta}_{rxn} = 437.5 KJmol^{-1}$

**CHM 003: PHYSICAL CHEMISTRY**

**5.a**

**i)** The reactivity of the group II elements of magnesium, calcium and barium with cold water increases down the group. Magnesium reacts so slowly to produce small bubble of hydrogen gas and magnesium hydroxide which is only slightly soluble.

$$Mg_{(s)} + H_2O_{(l)} \longrightarrow Mg(OH_2) + H_{2(g)}$$

Calcium reacts steadily with cold water to produce calcium hydroxide and hydrogen gas

$$Ca_{(s)} + H_2O_{(l)} \longrightarrow Ca(OH)_2 + H_{2(g)}$$

Barium reacts vigorously with cold water to produce barium hydroxide and hydrogen similar in strength to that of sodium in group I

$$Ba_{(s)} + H_2O \longrightarrow Ba(OH)_2 + H_{2(g)}$$

**ii)** $Mg_{(s)} + H_2O_{(l)} \longrightarrow MgO_{(s)} + H_{2(g)}$

**5.b** Aluminium

Equation on reaction

$4Al_{(s)} + 3O_{2(g)} \longrightarrow 2Al_2O_3$

**5.c**

**i)** The first ionization energy is the energy required to remove one mole of the most loosely held electrons from one mole of gaseous atoms to produce one mole of gaseous ions each with charge $+1$.

**ii)** The ionization energy of nitrogen is greater than oxygen because $P - orbital$ of nitrogen is half – filled orbital and thus it is stable than $P - orbital$ of oxygen in which unpaired electrons. Stability is less. Therefore, it is hard to remove an

electron from the outermost shell of stable nitrogen.

**iii)**

**(1)** Carbon atoms bind to each other not only in straight chains but in complex branching like the branches of a tree.

**(2)** Carbon ability to form long carbon – to – carbon chains, so that there can be so many different carbon compounds.

**(3)** Carbon atoms can share not only a single electron with another atom to form single bond but it can also share two or three electrons, forming a double or triple bond.

**(4)** In carbon atoms, the collection of atoms and bond are the same but in a different geometrical arrangement within the molecule, makes a molecule with different shape and hence different properties.

**(5)** All of the electrons that are not being used to bond carbon atoms together into chains and rings can be used to form bonds with atoms of several other elements.

**6.a**

**i)** Complex ion is an ion in which a central metal cation is bonded by a dative bond to a number of anions or neutral molecules called ligand in excess of its oxidation number or valency.

**ii)** Ligand

**6.b**

**i)**

| *Complex ion* | *Co – ordination Number* |
|---|---|

| | |
|---|---|
| $[Cu(NH_3)_4(H_2O)_2]^{2+}$ | 6 |
| $[Cu(H_2NCH_2CH_2NH_2)_2]^{2+}$ | 2 |

**ii)** diaquabis(ethylenediamine) iron (ii) ion

**iii)** diamminetetrachloro nickelate (ii) ion

**iv)** triamminetriaqua chromium (iii) chloride

**v)** tris (ethylenediamine) cobalt (iii) sulphate

**6.c**

**i)** Beryllium tends to be different from the other elements in group 2. This is due to exceptionally small atomic and ionic ($Be^{2+}\ ion$) size, high ionization enthalpy high, electronegativity and absence of $d-\ orbital$ in its valence shell.

Anomalous properties of beryllium

**(1)** Beryllium is harder than other members of its group.

**(2)** Beryllium has higher melting and boiling point that other member of its group.

**(3)** Beryllium does not react with water even at high temperature.

**(4)** Beryllium forms covalent compounds whereas other members form ionic compounds because of covalent character, salts of beryllium are easily hydrolyzed.

**(5)** Beryllium oxide and hydroxide are amphoteric whereas oxides and hydroxides of other alkaline earth metals are basic.

$$BeO\ +\ 2HCl\ \longrightarrow\ BeCl_2\ +\ H_2O$$

$$BeO\ +\ 2NaCl\ \longrightarrow Na_2BeO_2\ +\ H_2O$$

**(6)** Carbides of Be are covalent where as carbides of other members are ionic.

**(7)** Beryllium carbides react with water to produce methane gas where as carbide of other alkaline earth metals give acetylene gas.

$$BeC\ +\ 4H_2O\ \longrightarrow Be_2(OH)_2\ +\ CH_4$$

$$CaC_2\ +\ 2H_2O\ \longrightarrow Ca\,(OH)\ +\ HC\ \equiv\ CH$$

**(8)**

**(a)** Beryllium does not exhibit coordination number of more that four since it valence shell has only four (4) orbitals, one (1) S and P. The remaining elements of this group can show a coordination number of six by making use of $d - orbitals$ in addition to $S$ and $P$ orbitals.

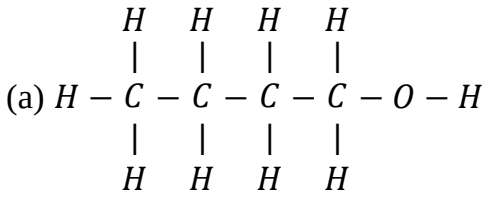
**(b)** Both carbonates and nitrates of the alkaline earth metal (group 2 elements) become more thermally stable down the group. The larger compounds further down require more heat than the lighter compounds in order to decompose. All the alkaline earth metal carbonates undergo thermal decomposition to the metal oxide and Carbon dioxide gas. Down the group, the carbonates require more heating to decompose while all the alkaline earth metal nitrate undergo thermal decomposition to the metal oxides, nitrogen dioxide and oxygen gas. Down the group, the nitrates require more heating to decompose.

**CHM 004: PHYSICAL CHEMISTRY**

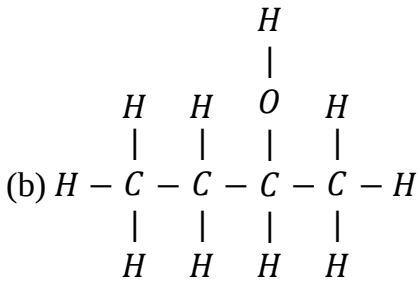
**7.a**

**i)** Isomerism is a phenomenon whereby two or more chemical compounds have the same molecular formula but different molecular structures due to differences in the arrangement of their atoms in space or the sequence of bonding of their atoms.

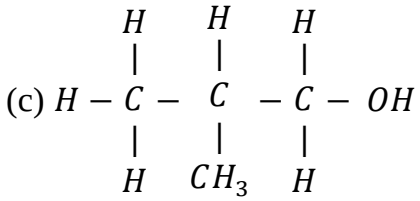
**ii)**

(a)


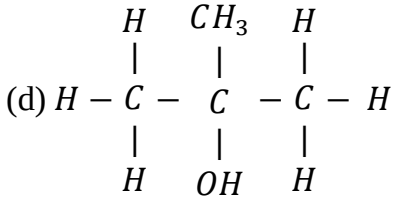
Butan – 1 - ol

(b)


Butan – 2 - ol

(c)


2 – methlypropan – 1 - ol

(d)


2 – methlypropan – 2 - ol

**iii)**

**(1)** $Butanoic\ acid$

**(2)** $Butan-2-one$

**(3)** $2-methylpropanoic\ acid$

**(4)** No reaction

**7.b**

**i)**

(1) Conversion of ethanoic acid to acid chloride

$$CH_3 - \overset{\overset{O}{\|}}{C} - OH \ + \ PCl_5 \longrightarrow \underbrace{CH_3 - \overset{\overset{O}{\|}}{C} - Cl}_{ethanoylchloride} \ + \ POCl_3 + HCl$$

(2) Conversion of ethanoic acid to acid antydride

Dehydration process by the action of Zinc oxide at $250°C$

$$\underbrace{H_3C - C - OH + HO - C - CH_3}_{Ethanoic\ acids} \xrightarrow[250°C]{ZnO} \underbrace{H_3C - \overset{\overset{O}{\|}}{C} - O - \overset{\overset{O}{\|}}{C} - CH_3 + H_2O}_{Ethanoic\ anhydride}$$

(3) Formation of ester from ethanoic acid

$Esterification\ reaction: Reaction\ of\ alkanoic\ acid\ with\ Alkanol$

$in\ the\ presence\ of\ a\ mineral\ acid\ (H^+)$

$to\ yield\ Alkyl\ alkanoates\ (ester)\ and\ water\ (H_2O)$

$$\underbrace{CH_3COOH}_{ethanoic\ acid} + \underbrace{C_2H_5OH}_{ethanol} \rightarrow \underbrace{CH_3COOC_2H_5}_{ethyl\ ethanoate} + H_2O$$

**7.c**

**i)** Ethanoic acid on heating with Sulphuric acid, $H_2SO_4$ or Phosphorus (v) oxide, $P_2O_5$ to produce acetic anhydride or ethanoic anhydride.

**ii)** $$H_2C - \underset{\underset{OH}{|}}{\overset{\overset{O}{\|}}{C}} + \underset{\underset{HO}{|}}{\overset{\overset{O}{\|}}{C}} - CH_3 \overset{H^+}{\rightleftharpoons} CH_3C - \underbrace{\overset{\overset{O}{\|}}{C} - O - \overset{\overset{O}{\|}}{C}}_{Ethanoic\ anhydride} - CH_3 + H_2O$$

Or

$$2CH_3COOH \xrightarrow[or\ P_2O_5]{H^+} (CH_3CO)_2O + H_2O$$

**iii)** Reaction of ethanoic acid with an ethanol and sulphuric acid, $H_2SO_4$

$$CH_3COOH + C_2H_5OH \xrightarrow[Acid\ catalyst]{} \underbrace{CH_3COOC_2H_3}_{Ester} + H_2O$$

Or

$$CH_3-\underset{\underset{O}{|}}{\overset{\overset{C}{\|}}{C}}+CH_3-\underset{\underset{H}{|}}{\overset{\overset{H}{|}}{C}}-OH \overset{H^+}{\rightleftharpoons} CH_3-\underbrace{\underset{\underset{O}{\|}}{\overset{\overset{C}{|}}{C}}-O}_{Ester}-CH_2CH_3+H_2O$$

**8.a**

**i)** Grignard reagents are organometallic compound that are formed when halo alkanes react with magnesium metal in the presence of ether to yield an alkyl $magnesium\ (II)\ halide\ or\ phenyl\ analogous$

**ii)** Preparation of Grignard reagent

i. $$\underbrace{RX}_{Haloalkane}+\underbrace{Mg}_{Magnessium}\overset{ether}{\rightleftharpoons}\underbrace{RMgX}_{\substack{alkyl\ magnessium\ (II)\ chloride\\(Grignard\ reagent)}}$$

ii. $$\underbrace{CH_3CH_2Cl}_{Chloroethane}+Mg\overset{ether}{\rightleftharpoons}\underbrace{CH_3CH_2MgCl}_{\substack{ethylmagnessium(II)chloride\\(Grignard\ reagent)}}$$

**iii)** Preparation of the reaction

$$\underbrace{CH_3CH_2\overset{\overset{O}{\|}}{C}-H}_{Propanol}+\underbrace{CH_3CH_2MgBr}_{Ethylmagnessium(II)bromide}\rightarrow$$

$$CH_3CH_2\underset{\underset{H}{|}}{\overset{\overset{CH_3CH_2}{|}}{C}}-OMgBr\overset{HBr}{\rightleftharpoons}$$